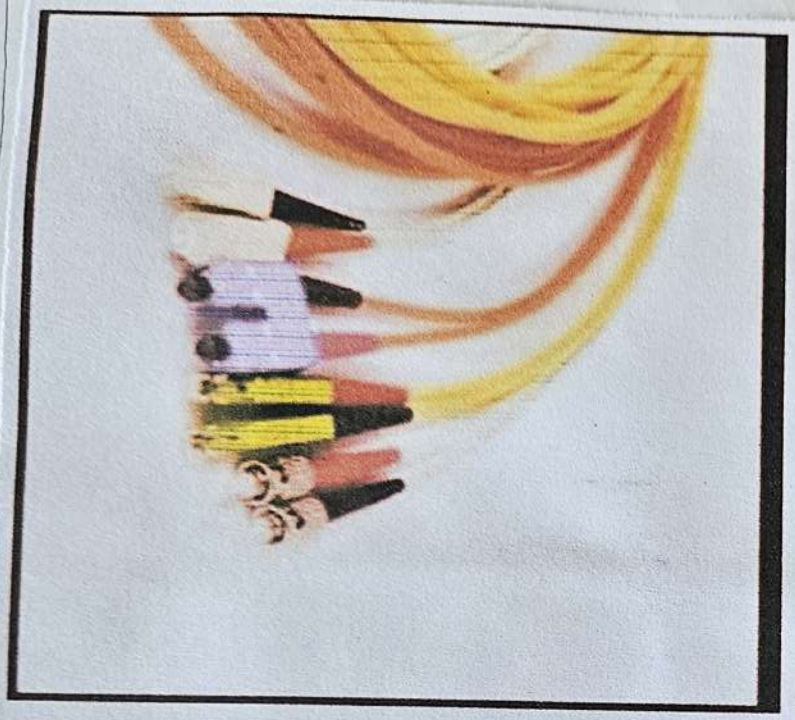
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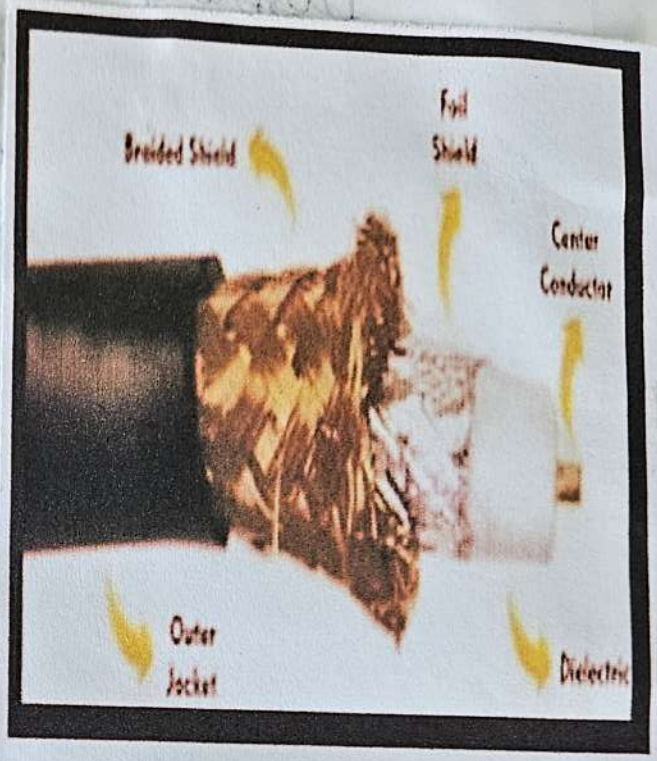
STUDY OF DIFFERENT TYPES OF NETWORK CABLES.

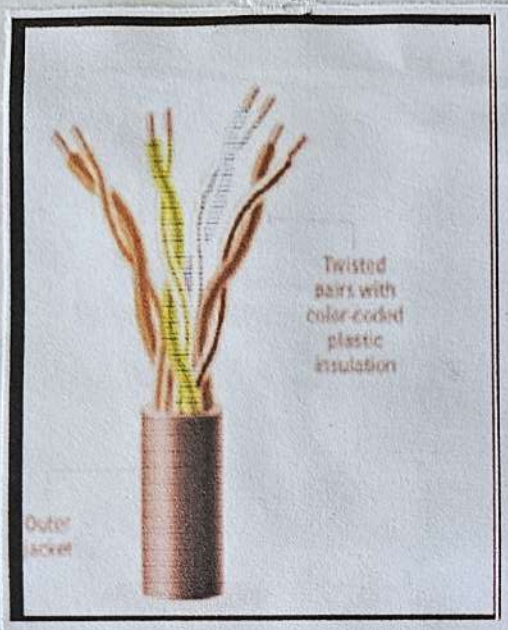
Aim:

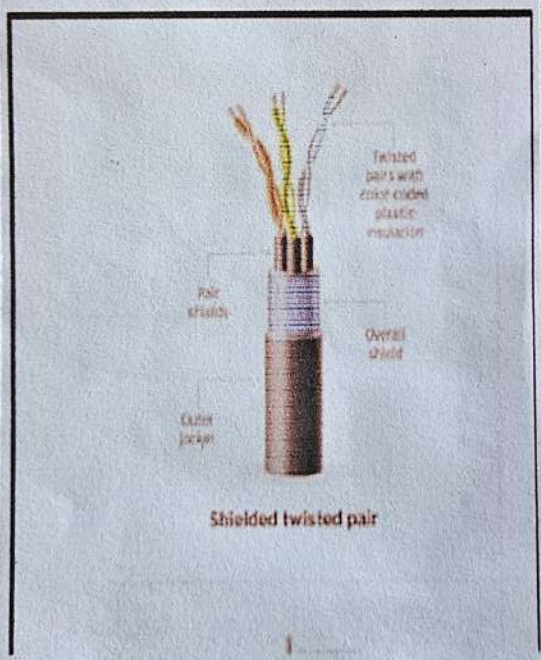
To understand different types of network cables.

Cable Type	Category	Max data Transmission	Advantages/ disadvantages	Application / use
UTP	Category 3	10bps	Advantages: 1. Cheaper in cost 2. Easy to install as they have smaller diameter Disadvantages: 1. More prove to EMI.	10 Base ethernet
	Category 5	upto 100 mbps		Fast Ethernet, Gigabit Ethernet
	Category 5e	1GPs		Fast Ethernet, Gigabit Ethernet
STP	Category 6, 6a	10GPS	Advantages: 1. Shielded 2. Faster than UTP 3. Less susceptible Disadvantages: 1. Expensive 2. Greater installation Effort	Gigabit Ethernet, 10G Ethernet (55m) widely used in data centres.
SSTP	Category 7	10GPS		Gigabit Ethernet 10G Ethernet (100m)
Coaxial cable	RG-6 RG-59 RG-11	10-100Mbps	Advantages: 1. High bandwidth 2. Immune to interference 3. Low loss bandwidth 4. Versatile Disadvantage: 1. Costly 2. Size is bulky	Speed of signal is 500m. Television network High speed internet connections
Fibre optics cable	Single mode Multi mode	100GPS	Advantages: 1. High speed 2. High Bandwidth 3. High security 4. Long Distance Disadvantage: 1. Expensive 2. Require skilled installer	Maximum distance of fibre optical cable is around 100m.

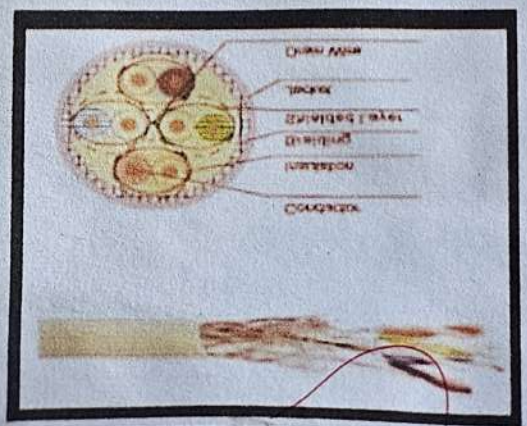



Braided Shield
Foil Shield
Center Conductor
Outer Jacket
Dielectric





Shielded twisted pair



Result:

Thus, we have learned the different types of network cables for computer networks.